A Fixed Confidence Gate for Selective Classification

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Abstract

We evaluate a fixed confidence gate for selective classification using two synthetic fixtures. The gate abstains below a specified confidence threshold. We report accuracy among accepted predictions together with coverage. Conditional accuracy rises on one dataset and falls on the other, without establishing an overall accuracy improvement. A selective-risk range bound requires bounded loss and positive coverage. Uncertainty, runtime, and distribution shift remain unevaluated.

1. Introduction

Selective classification allows a model to abstain from uncertain predictions. We evaluate a fixed confidence gate using synthetic fixtures. Because the gate selects which predictions are evaluated, we report coverage alongside accuracy among accepted predictions.

2. Method

Gate uses a fixed classifier and accepts a prediction when its maximum class probability is at least 0.8; otherwise, it abstains. Always uses the same classifier and accepts every prediction. Neither method retrains the model. For a fixed classifier h and acceptance function $a(x) \in \{0, 1\}$, assume $0 \leq \ell(h(x), y) \leq 1$ and positive coverage $C = \mathbb{E}[a(X)] > 0$. Selective risk is

$$R_{\text{sel}} = \frac{\mathbb{E}[a(X)\ell(h(X), Y)]}{C}. \tag{1}$$

Theorem 1 *Under these assumptions,* $0 \leq R_{\text{sel}} \leq 1$.

This range bound provides neither a generalization guarantee nor a comparison between Gate and Always. Appendix A gives the full proof.

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3. Synthetic Evaluation

Alpha and Beta each contain 1000 synthetic samples. Both methods use the same model and 5 evaluation seeds; results are seed means, with no variance estimates.

Table 1. Synthetic fixture values. Always accuracy uses all samples; Gate accuracy is conditional on acceptance. These values are not research findings.

Dataset	Always (%)	Gate (%)	Coverage (%)
Alpha	80.0	84.0	60.0
Beta	76.0	75.5	58.0

Table 1 shows a higher conditional accuracy on Alpha and a lower conditional accuracy on Beta. Different evaluation populations prevent interpreting these differences as an overall accuracy improvement. No significance tests, runtime measurements, or distribution-shift evaluations were performed.

4. Limitations and Conclusion

The synthetic evaluation compares a fixed gate with a single baseline. It does not establish state-of-the-art performance or a guarantee of lower risk. The classifier and data-generation details remain missing and are required for reproducibility. Conditional accuracy and coverage describe the accepted population; deployment benefits remain untested. The source template follows the ICML author guidelines (for, 2026).

Impact Statement

This manuscript is a synthetic fixture for evaluating writing and template migration skills; its values are not research findings. Reporting coverage with conditional accuracy clarifies how abstention changes the evaluation population. The fixture supports no claim of improved deployment outcomes, social benefit, universal applicability, or faster execution.

References

Icml 2026 author instructions. https://icml.cc/Conferences/2026/AuthorInstructions, 2026.

A. Proof of the Range Bound

Since $0 \leq a(X)\ell(h(X), Y) \leq a(X)$, taking expectations gives $0 \leq \mathbb{E}[a(X)\ell(h(X), Y)] \leq C$. Division by $C > 0$ proves Theorem 1. This argument applies to the specified distribution; it does not establish a bound under distribution shift.